

Short Activity: Using a DAG to Discuss Design-Based vs. Conditioning-Based Approaches

Duration: 10–15 minutes

Objective

- Understand how a Directed Acyclic Graph (DAG) can illustrate potential causal relationships in political research.
- Encourage a conceptual comparison of *design-based* and *conditioning-based* approaches to address confounding factors.

Context

Imagine we have a research question: Does a local official's personal relationship with central leaders (X) affect the level of infrastructure investment in that local district (Y)?

We suspect there are other factors—for instance, the region's economic needs, the official's personal reputation, or party affiliations—that might influence both X and Y.

Activity Steps

1. Form Small Groups (2–4 students each)

Distribute a blank piece of paper to each group. Ask them to think about the potential pathways or variables (nodes) that could affect or be affected by X and Y .

2. Draw a Simple DAG

Each group sketches a DAG with arrows indicating causal directions. For example:

- X (personal tie to central leaders) $\rightarrow Y$ (infrastructure investment).

- A confounder (e.g., “regional demand”) influencing both X and Y .
- Possibly a mediator (e.g., “official’s competence”) on the path from X to Y .

3. Brainstorm Methods to Estimate the Causal Effect of X on Y

Introduce the following two conceptual strategies and let each group discuss them briefly:

- **Design-Based Approach:** Seek an external or random event that makes X vary in a quasi-random manner (e.g., a natural experiment, an unexpected leadership reshuffle, or a boundary rule).
- **Conditioning-Based Approach:** Control or adjust for confounders in the analysis (e.g., including control variables in regression, matching observations, or fixed effects).

4. Group Share-Out

Have each group briefly present:

- Their DAG and what paths or variables they considered critical.
- Which approach (design-based vs. conditioning-based) might be more *feasible* and why, given the scenario.

Wrap-up Discussion

Conclude by emphasizing:

- DAGs are conceptual tools for *organizing* our thinking about which variables might confound the relationship of interest.
- A **design-based approach** aims to exploit external variation or quasi-experimental conditions to rule out confounding, whereas a **conditioning-based approach** tries to *control* for observed confounders (in regression, matching, etc.).
- Real-world data availability and feasibility concerns often guide which approach is chosen.